B. Maximum Multiple Sum

time limit per test 1 second
memory limit per test 256 megabytes
input standard input
output standard output

Rating: 800 | Tags: brute force, math, number theory | https://codeforces.com/problemset/problem/1985/B

Given an integer n, find an integer x such that:

- $2 \le x \le n$.
- The sum of multiples of x that are less than or equal to n is maximized. Formally, $x + 2x + 3x + \dots + kx$ where $kx \le n$ is maximized over all possible values of x.

Input

The first line contains t ($1 \le t \le 100$) — the number of test cases.

Each test case contains a single integer n ($2 \le n \le 100$).

Output

For each test case, output an integer, the optimal value of x. It can be shown there is only one unique answer.

Example

Input
2 3 15

Output
3 2

Note

For $n = 3$, the possible values of x are 2 and 3. The sum of all multiples of 2 less than or equal to n is just 2, and the sum of all multiples of 3 less than or equal to n is 3. Therefore, 3 is the optimal value of x.

For $n = 15$, the optimal value of x is 2. The sum of all multiples of 2 less than or equal to n is $2 + 4 + 6 + 8 + 10 + 12 + 14 = 56$, which can be proven to be the maximal over all other possible values of x.